

Electronic Companion

Service Contracts with Limited Memory:
Participation, Randomization, and Exact Design

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Reading this companion

The main article is organized around restricted-memory renewal design. This companion retains the broader compact-response, tight-size, piecewise-quadratic, behavioral-minimization, dispersion, and shared-table theory with its original hypotheses. Paper-specific bounded-overflow proofs and current computational methods appear in the article appendices, with their measurement tables after the article references. No implication is transferred from the scalar no-switching architecture to the older general switching or coupled-commitment model without its stated assumptions.

References whose numbers begin with H point to the exact preceding R36 readers in the preservation archive, not to a missing current section. Older experiment narratives, unfavorable findings, and revision concordances remain there and in their unchanged source modules. The current source-preservation map identifies each location; no historical theorem or data file is deleted.

EC.1 Service Agreements and Their Laminar Allocation Form

EC.1.1 Public primitives and accepted continuations

Let $G = (\mathcal{V}, \mathcal{E})$ be a finite, rooted, acyclic public graph. Vertices include date; all are reachable. Write $N = |\mathcal{V}|$, $M = |\mathcal{E}|$, and let $p_{vw} > 0$ be the exogenous probability on edge (v, w) , with probabilities summing to one at a nonterminal vertex. The discount factor is $0 < \beta \leq 1$. An occurrence h is a root-to-vertex history in the finite unfolding $\mathcal{T}$; its public vertex is $v(h)$ and its discounted reach weight is $w_h = \beta^{t(h)} \Pr(h) > 0$. Write $j \succeq h$ for a descendant, including h , and $H = |\mathcal{T}|$.

At occurrence h of v , the provider chooses a tier $x_h \in [l_v, h_v] \subseteq [0, 1]$. The payment coefficient $a_v > 0$ is fixed, and the operating reward is $g_v(x) = r_v x - q_v x^2/2$, with $q_v > 0$. Customer utility is a publicly specified service-benefit term minus $a_v x$. For example, fixed stock s_z yields

$$u(z, s_z, x) = v \mathbb{E} \min(s_z, D_z) - a_z x, \quad a_z = r_z^{\text{premium}} - \kappa \mathbb{E}(D_z - s_z)^+ > 0. \quad (\text{EC.1})$$

Stock may be stipulated, or identified by the service and cost commitments in the retained stock-identification results. The customer can leave before each modeled review; the provider commits to its contingent protocol. Retaining the customer requires the conditional service benefit minus remaining payment to exceed the public

outside continuation. Assume their difference is a vertex-sufficient cap B_v . The accepted optimization is

$$\begin{aligned}
 J(b_0) = \max_x \quad & \sum_{h \in \mathcal{T}} w_h g_{v(h)}(x_h) \\
 \text{subject to} \quad & Y_h := \sum_{j \succeq h} \frac{w_j}{w_h} a_{v(j)} x_j \leq B_{v(h)} \quad (h \in \mathcal{T}), \quad Y_0 = b_0, \\
 & l_{v(h)} \leq x_h \leq h_{v(h)}.
 \end{aligned} \tag{EC.2}$$

The root has a cap as well as the equality promise. No privately informed reports, endogenous termination choice, or policy-dependent transition probabilities are part of this institution. The no-switching restriction is essential to the scalar-price result; the retained general model continues to allow switching and is not replaced by (EC.2).

The feasible remaining-payment interval is computed without unfolding:

$$L_v = a_v l_v + \beta \sum_w p_{vw} L_w, \quad U_v = \min \left\{ B_v, a_v h_v + \beta \sum_w p_{vw} U_w \right\}. \tag{EC.3}$$

A subtree is feasible precisely when its successors are feasible and $L_v \leq U_v$. Throughout, $b_0 \in [L_0, U_0] \subseteq (-\infty, B_0]$. Convexity fills every point between the two extreme payments. Compactness gives an optimum and strict concavity gives a unique occurrence-indexed action vector.

EC.1.2 An exact mapping, not an analogy

Proposition EC.1.1 (Laminar allocation equivalence). *Define $z_h = w_h a_{v(h)} x_h$. Problem (EC.2) is exactly the separable concave allocation problem*

$$\begin{aligned}
 \max_z \quad & \sum_h \left[\frac{r_{v(h)}}{a_{v(h)}} z_h - \frac{q_{v(h)}}{2w_h a_{v(h)}^2} z_h^2 \right], \\
 \sum_{j \succeq h} z_j & \leq w_h B_{v(h)}, \quad \sum_j z_j = b_0, \\
 w_h a_{v(h)} l_{v(h)} & \leq z_h \leq w_h a_{v(h)} h_{v(h)}.
 \end{aligned} \tag{EC.4}$$

The constrained descendant sets are laminar. The change of variables is invertible and preserves objective values and optimal policies.

Proof. Substitution gives each term and row of (EC.4). Two rooted descendant sets in a tree are either disjoint or one contains the other. Positivity of $w_h a_{v(h)}$ makes the coordinate map bijective and preserves every inequality direction. $\square$

Thus the expanded primal lies in the tree-constrained allocation lineage of Mjelde (1983) and Tang (1990). Tang reports a reduction to one-variable convex subproblems; that result is part of the inherited optimization structure, not a competitor we relabel. The chain-nested allocation results of Vidal et al. (2016);

Schoot Uiterkamp et al. (2021); Wu et al. (2021) concern related but different input structures. A complexity bound for an ascending chain cannot simply be assigned to arbitrary branching descendant sets.

After subtracting lower allocations, the packing region is a laminar polymatroid, and the exact total selects a base of its truncation (Electronic Companion, Section EC.7). This connects the model to Federgruen and Groenevelt (1986) and Groenevelt (1991). Harks et al. (2018) further show that sensitivity analysis for separable optimization over integral polymatroid bases is a substantial literature in its own right. Their decision set is integral and their perturbations concern costs and polymatroid parameters; our theorem concerns a continuous quadratic accepted-service model and a complete response in one dual price. We cite that work as sensitivity lineage, not as a result to be rediscovered.

EC.1.3 What remains after memoization and generic parametric optimization

An occurrence-indexed marginal reduction may represent subtree payment in the unnormalized units z . Its entire response is the occurrence weight times a common conditional response, proved below. Dividing by that weight makes occurrences of the same public vertex identical. Memoizing those normalized responses therefore gives our graph recursion. There is no additional stochastic obstruction or different first-order condition. We do not assert that scalar marginal allocation, subtree reduction, or memoization originated here.

The closest modern parametric comparison is Klimm and Warde (2021), who construct output-polynomial algorithms for separable, continuous, piecewise-quadratic parametric minimum-cost flow. Their parameter scales network demands and their feasible set is a flow polyhedron; our parameter is the price conjugate to an accepted-service payment and the unfolded feasible set is laminar allocation. A laminar problem may itself be reformulated before a parametric method is applied, so we make no priority claim for generic piecewise-affine solution paths or for output-sensitive parametric optimization. The contribution claimed here is narrower and quantitative: after quotienting repeated occurrences to the supplied public graph, all response changes are generated by local clipping events and at most one cap barrier per public vertex. This gives an at-most- $3N$ global knot universe, a matching $\Omega(N)$ family, a matching $\Omega(N^2)$ total-response family, and an explicit rational bit bound measured in the compact graph input. These conclusions are proved directly in Section EC.2.

This positioning answers five distinct questions. First, the complete response is a special structured parametric convex solution map, so pathwise parametricity itself is not claimed. Second, an unfolded laminar algorithm can compute the same path, and normalized memoization yields the same recurrence. Third, the compact quotient makes the response-size bound depend on public vertices rather than occurrence histories. Fourth, the global knot and total-storage orders are shown below to be tight for polynomial-size rational instances. Fifth, the bit bound is for the explicit coefficient-event constructor and includes exact sorting and comparison cost; it is not a generic encoding theorem for every parametric solver.

EC.1.4 Behavioral minimization and policy-graph lineage

The reached execution object is also connected to classical minimization. Bean et al. (1987) study aggregation in dynamic programming; Givan et al. (2003) develop state equivalence and model minimization for MDPs; and Peyrière (2023) gives a minimum-state equivalent Moore-machine construction. Our backward equality of current outputs and successor classes is an instance of that classical deterministic behavioral-minimization principle. Section EC.4 therefore states the reached price system as an output transducer first, applies standard minimization, and then isolates the contract-specific results: finite reachability from continuous controls, ordered contiguous behavior classes, separate writable/read-only accounting, and a linear contractual lower-bound family.

Promised continuation utility is a standard recursive state in dynamic contracting (Spear and Srivastava, 1987; Thomas and Worrall, 1988; Chen et al., 2020). Zhang (2012) is especially relevant because it uses finite policy graphs to represent continuation contracts and payoff frontiers in an infinite-horizon adverse-selection model with Markov information. That policy graph is an approximation device for a different information and incentive problem. Our public-information model is finite horizon, has no private reports, and compiles an exact transducer for a fixed root promise. The comparison is nevertheless conceptually close: the present result can be read as an exact policy-graph compression theorem for this scalar public-information subclass.

Policy-graph decomposition and stochastic decomposition already avoid naive scenario enumeration (Pereira and Pinto, 1991; Girardeau et al., 2015; Dowson, 2020). We compare an explicit primal promise-grid method on the public graph rather than treat the mere use of a graph as a contribution. Our exact closure depends on one scalar additive commitment and separable quadratics; general convex policy-graph decomposition has a broader domain. The supporting cuts for fixed tables are also instances of standard dual decomposition (Benders, 1962; Boyd and Vandenberghe, 2004), not a new general finite-convergence theorem.

Input-size convention. An explicit-tree representation has H local records and must at least expose those records to an occurrence-indexed implementation. Our compact representation has N vertices and M edges. This is an input/output comparison, not a lower bound against every algorithm that accepts a compact graph. The independent tree code is retained as an exact cross-check, not as a stand-in for the strongest classical or modern specialized solver.

EC.2 A Tight Quantitative Quotient of the History Tree

EC.2.1 Conditional responses and cap reflection

For an occurrence h of v , let $H_h(\eta)$ be the maximum conditional reward minus ηY_h , imposing all subtree caps but not an exact payment equality at h . Let $H_h^{\text{pre}}(\eta)$ remove only the cap at h , retaining every successor cap. Conditional rewards use weights w_j/w_h . Define $R_h(\eta)$ as the payment of the unique maximizer of $H_h(\eta)$; use $d_h(\eta)$ for the pre-cap payment. These definitions separate a removed current cap from the constrained continuation problems below it.

Theorem EC.2.1 (Occurrence-to-vertex quotient and finite response closure). *Under the assumptions of Section EC.1, occurrences h, h' of a public vertex v with the same incoming price η have identical normalized optimal continuation plans. In particular, $H_h = H_{h'} =: H_v$, $H_h^{\text{pre}} = H_{h'}^{\text{pre}} =: H_v^{\text{pre}}$, and $R_h = R_{h'} =: R_v$. The following backward construction computes the common payment responses and optimal policies:*

$$x_v^0(\eta) = [(r_v - a_v\eta)/q_v]_{[l_v, h_v]}, \quad (\text{EC.5})$$

$$d_v(\eta) = a_v x_v^0(\eta) + \beta \sum_w p_{vw} R_w(\eta), \quad (\text{EC.6})$$

$$R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (\text{EC.7})$$

Write $d_v^- = a_v h_v + \beta \sum_w p_{vw} U_w$ for the finite constant on the left tail of d_v . If $B_v \geq d_v^-$, assign no barrier at v . Otherwise let α_v be the leftmost finite solution of $d_v(\alpha_v) = B_v$. The optimal local action and the incoming price sent to every successor are determined by

$$s_v = \max\{\eta, \alpha_v\}, \quad x_v = x_v^0(s_v), \quad (\text{EC.8})$$

where a missing barrier leaves $s_v = \eta$.

Every R_v is continuous, nonincreasing, piecewise affine with constant tails and range $[L_v, U_v]$. The global universe of distinct response knots has size at most $3N$. Complete response tables can be constructed in

$$A = O((N + M)N \log(N + 1)) \quad (\text{EC.9})$$

exact rational arithmetic operations and exact comparisons, using $O(N^2 + M)$ scalar storage. For a fixed root promise, choose any finite η_0 with $R_0(\eta_0) = b_0$. Execution uses at most $N + 1$ distinct incoming-price labels, drawn from η_0 and the finite cap barriers. The construction does not enumerate H occurrences.

Proof. The subtree below an occurrence is determined by its displayed vertex and subsequent public transitions. The map replacing the initial occurrence by another occurrence of the same vertex preserves all suffix paths, relative probabilities, discounts, local rewards, bounds, and caps. It is consequently a bijection of the normalized feasible problems at any common η . Strict concavity makes their maximizing action vectors agree. This proves the quotient independently of the algorithm.

Remove the current cap. Separability gives (EC.5) and independent successor problems at the same incoming price, proving (EC.6). Their unique maximizers have continuous payment responses. Backward induction starts with a clipped affine local payment and preserves continuity, monotonicity, and affine pieces under weighted summation. Its two tails are the extreme attainable payments before the current cap.

When the current cap is redundant, this pre-cap maximizer is feasible. Otherwise feasibility implies that B_v belongs to the range of d_v . The clipped finite response reaches its tails at finite prices, so a finite leftmost crossing exists. For $\eta < \alpha_v$, the pre-cap optimum at α_v has payment B_v . Let z^* be that optimum and z any policy satisfying the current cap. Writing G for conditional reward and Y for payment gives

$$G(z) - \eta Y(z) \leq G(z^*) - \alpha_v B_v + (\alpha_v - \eta) Y(z) \leq G(z^*) - \eta B_v.$$

For $\eta \geq \alpha_v$, the pre-cap optimizer already satisfies the cap. These two cases prove reflection, (EC.7), and its implementing policy. If the payment is constant between two prices, the two optimality inequalities imply that either unique optimizer is also optimal at the other price. Thus the policy is unchanged on an equality plateau even when the supporting dual price is not unique. This also covers a cap at the minimum feasible payment and fixed local intervals.

Local clipping contributes at most two knots. Summing child responses introduces no knot outside their union. Capping can remove knots and introduces at most one crossing. Inducting over distinct *vertices*, rather than occurrences, gives a global union of at most $3N$. Each of the N responses can still have order N segments, so total response storage is quadratic, not linear. Directly merging weighted coefficient changes at sorted knots gives (EC.9); the graph takes an additional $O(N + M)$ records.

Finally, each executed price is the maximum of the root price and the barriers along the realized path. There are at most $N + 1$ such numerical labels for this fixed root query, and at most $N(N + 1)$ reachable vertex–price pairs. The root maximizer at η_0 has precisely b_0 , so it also maximizes reward among policies satisfying the equality. $\square$

The unnormalized response at occurrence h in Proposition EC.1.1 is $w_h R_{v(h)}(\eta)$. This identity is the explicit link to an occurrence-indexed tree reduction. A normalized memoized implementation enjoys the same quotient recurrence; it is not an algorithmic competitor that our implementation is claimed to dominate.

EC.2.2 Matching lower bounds for the response representation

The preceding upper bounds are not artifacts of a loose counting argument.

Theorem EC.2.2 (Worst-case tight response size). *For every integer $n \geq 1$, there is an n -vertex rational instance with $M = n - 1$ for which the global response-knot universe has exactly $2n$ distinct knots and the total number of affine response segments is exactly $n^2 + 2n$. Consequently the $O(N)$ global-knot bound and the $O(N^2 + M)$ complete-response storage bound in Theorem EC.2.1 are worst-case tight up to constant factors, even with no binding continuation caps.*

Proof. Take the deterministic chain $1 \rightarrow 2 \rightarrow \dots \rightarrow n$, with $\beta = 1$, edge probability one, $a_i = q_i = 1$, $l_i = 0$, $h_i = 1$, and $r_i = 2i$. Set the continuation cap at vertex i to

$$B_i = n - i + 1.$$

This cap equals the maximum feasible remaining payment, so it is redundant and creates no barrier. The response at vertex i is therefore

$$R_i(\eta) = \sum_{j=i}^n [2j - \eta]_{[0,1]}. \quad (\text{EC.10})$$

For every j , the local summand has knots $2j - 1$ and $2j$. These $2n$ numbers are distinct. At each such number the slope of exactly one summand changes by one in magnitude, so no knot cancels in any ancestor response that contains that summand. Hence R_i has exactly $2(n - i + 1)$ knots and $2(n - i + 1) + 1$ affine

segments. Summing over i gives

$$\sum_{i=1}^n [2(n-i+1) + 1] = \sum_{k=1}^n (2k+1) = n^2 + 2n.$$

All primitives are integers with $O(\log n)$ bits, so the family has polynomial encoding length. The claimed asymptotic lower bounds follow. $\square$

The theorem separates two notions that can otherwise be conflated. There are only linearly many distinct knot *locations*, but a downstream knot can appear in the response table of every ancestor, producing a quadratic number of stored affine pieces. The dense computational family in Section H.10 is an empirical counterpart of this analytic construction; the lower bound itself does not depend on computation.

Corollary EC.2.3 (Several coordinates under one scalar commitment). *With d_v separable local coordinates, positive payment coefficients, positive quadratic coefficients, and a product interval, let $D = \sum_v d_v$. The knot universe has size at most $2D + N$, response storage is $O(N(2D + N) + M)$, and the incoming-price alphabet still has at most $N + 1$ elements for a fixed query.*

Proof. There are two clipping knots per coordinate, but only one scalar cap barrier per vertex. Clipping knots describe how a local action responds to a given price; they are not additional prices written into the running record. The common commitment price and cap reflection are unchanged. $\square$

Independent commitments can be separated only when rewards and feasible sets also separate. Coupled vector promises, nonseparable quadratic terms, and intertemporal switching do not satisfy the scalar reflection argument. Their retained primal promise-state and convex formulations remain applicable.

EC.2.3 Rational bit complexity

An arithmetic count alone does not establish polynomial time on binary inputs. Let L be the total binary encoding length of all rational primitives, graph indices, and the rational root query, with rationals in lowest terms. The following statement is specifically for the coefficient-event construction used in Theorem EC.2.1. Stored rational coefficients, event locations, and crossings are reduced to lowest terms after each exact arithmetic update. The common denominators used in the proof below are bounding devices; they are not materialized as a giant implementation denominator.

Theorem EC.2.4 (Polynomial rational preprocessing for the coefficient-event compiler). *All stored response slopes, intercepts, finite knots, cap barriers, and a returned finite root price can be represented with*

$$S = O(NL + \log(N + M + 1)) \tag{EC.11}$$

bits per reduced numerator and denominator. The event merges, sorting, exact comparisons, coefficient arithmetic, cap-crossing calculations, and one root inversion together admit the conservative bit-operation bound

$$O((N + M)N \log(N + 1) S^3) = O(AS^3). \tag{EC.12}$$

The read-only response representation occupies $O((N^2 + M)S)$ bits. These are polynomial bounds for rational inputs to this explicit compiler, not a strongly polynomial claim and not a generic bit bound for every parametric optimization algorithm.

Proof. Fix an open interval between global knots. At a vertex whose current cap binds on that interval, the payment response is the constant B_v . Otherwise its coefficients are a local affine payment plus the discounted, probability-weighted coefficients at successors. The possible nonzero local sources are

$$B_v, \quad a_v l_v, \quad a_v h_v, \quad a_v r_v / q_v, \quad -a_v^2 / q_v. \quad (\text{EC.13})$$

Expanding either coefficient through the acyclic graph therefore gives a sum over directed paths of one source multiplied by edge probabilities and a power of β . Expansion stops at a capped vertex. No path repeats an edge, and its length is at most $N - 1$.

Let D_0 be the product of all positive denominators and the absolute values of all nonzero numerators appearing in the rational primitive input. Its logarithm is $O(L)$. Every path-term denominator divides D_0^{N+4} ; this exponent covers repeated discounting, each edge probability, and the products and divisions in (EC.13). Thus a conceptual common denominator has $O(NL)$ bits even though the number of paths may be exponential. The sum of nonnegative discounted path weights to source occurrences is at most N , because it is bounded by the expected number of visited vertices. The absolute value of a source is at most $2^{O(L)}$. Multiplying by the conceptual denominator therefore bounds every reduced stored numerator and denominator by (EC.11).

The actual event compiler does not build D_0^{N+4} . It adds and multiplies ordinary reduced rationals while merging sorted coefficient events. Intermediate stored partial sums have the same path-product denominator structure and differ by at most a polynomial factor in $N + M$ in magnitude. A new cap crossing has form $(B_v - c)/m$ on a nonconstant line $m\eta + c$; local clipping knots and a root inverse are ratios of a constant number of bounded-height rationals. Their reduced heights also satisfy (EC.11).

Sorting and merging are part of the operation count in (EC.9). Every exact order test is a cross-product comparison of $O(S)$ -bit integers. Rational addition, multiplication, division, comparison, and gcd reduction can all be bounded conservatively by $O(S^3)$ bit operations. Multiplying that cost by the full arithmetic-and-comparison count proves (EC.12). Counting stored coefficients, knot locations, and graph entries proves the storage claim. $\square$

The coefficient height above is not the bit length of a completely printed primal–dual certificate. Aggregate execution rewards and certificate totals can combine different realized denominators and have a larger, still polynomial, height. The Electronic Companion reports that separate aggregate bound and the implementation records coefficient precision separately from certificate precision. This distinction is part of the memory accounting below.

Corollary EC.2.5 (Amortized exact root inversion). *After the complete response family is compiled, a new feasible root promise can be inverted on the stored root response using $O(\log(N + 1))$ exact comparisons and a constant number of rational arithmetic operations. This is an inversion bound only: emitting and auditing the reached execution machine for the new query is a separate operation.*

Proof. The root response has at most $3N$ globally indexed knots. Binary search locates the constant or affine segment containing the requested payment; one affine solve, or selection of a finite point on a constant tail, returns a valid root price. $\square$

EC.2.4 A compact exact certificate

For each reached (v, η) , propagate discounted occupation weight w , effective price s , action x , and multipliers $\chi = s - \eta$, $\ell, u \geq 0$. They satisfy

$$r_v - q_v x - a_v s + \ell - u = 0, \quad \ell(x - l_v) = u(h_v - x) = \chi(B_v - R_v(\eta)) = 0. \quad (\text{EC.14})$$

Weight $\beta w p_{vw}$ moves to (w, s) . A backward conditional-payment calculation verifies every cap; forward flow verifies occupation weights and the root promise. The resulting dual upper bound is

$$\eta_0 b_0 + \sum_{(v, \eta)} w \left[\chi B_v + u h_v - \ell l_v + \frac{(r_v - a_v s + \ell - u)^2}{2q_v} \right]. \quad (\text{EC.15})$$

Ancestor prices telescope under the flow equations, and (EC.14) makes (EC.15) equal the implemented reward. These rational checks certify a particular optimum; verifying entire response-curve identities is a separate implementation test. Neither kind of code execution replaces the mathematical proof above.

EC.3 Piecewise-Quadratic Rewards and Shared Response Representations

The quadratic model identifies the basic mechanism, but a single curvature need not describe all service tiers. A change in staffing intensity can change marginal cost, and crossing a service threshold can produce a downward jump in marginal reward. Both features preserve exact closure. The relevant primitive is a finite, continuous, nonincreasing local price response, not a globally constant curvature.

EC.3.1 Exact closure with changing curvature and marginal jumps

Replace the local reward in Section EC.1 by a continuous, strictly concave piecewise-quadratic function. More generally, vertex v may have d_v separable coordinates on a product interval, all contributing to the same scalar payment with coefficients $a_{vi} > 0$. For a nonfixed coordinate, let

$$l_{vi} = t_{vi,0} < t_{vi,1} < \dots < t_{vi,m_{vi}} = h_{vi}, \quad g_{vi}(x) = c_{vij} + r_{vij}x - \frac{1}{2}q_{vij}x^2 \quad (t_{vi,j-1} \leq x \leq t_{vi,j}),$$

where $q_{vij} > 0$. Adjacent formulas agree in value; the left derivative at a join is at least the right derivative. All breakpoints and coefficients are rational. A fixed coordinate contributes a constant and is recorded separately. Set $D = \sum_v d_v$, $J = \sum_{vi} m_{vi}$ over nonfixed coordinates, and $Q = 2J + N$. The transition, scalar commitment, and public cap assumptions remain those of Section EC.1.

Theorem EC.3.1 (Piecewise-quadratic quotient with a primitive event budget). *The local price response is*

$$\phi_{vi}(\eta) = \arg \max_{[l_{vi}, h_{vi}]} \{g_{vi}(x) - a_{vi}\eta x\}.$$

The occurrence quotient, payment recursion, and cap reflection remain exact, with

$$d_v(\eta) = \sum_i a_{vi} \phi_{vi}(\eta) + \beta \sum_w p_{vw} R_w(\eta), \quad R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (\text{EC.16})$$

Every response is continuous and piecewise affine. At most Q globally distinct response knots occur. Explicit response tables use

$$O(NQ + J + D + M)$$

rational scalars and can be constructed using

$$A_{pq} = O((N + M)Q \log(Q + 1) + (J + D) \log(J + D + 2)) \quad (\text{EC.17})$$

exact arithmetic operations and comparisons. If all joins are differentiable, $2J$ in the knot budget can be replaced by $\sum_{vi} (m_{vi} + 1)$ over nonfixed coordinates.

For total binary input length L , a coefficient-event implementation with reduction after each rational update has response-coefficient and event height

$$S_{pq} = O((N + 1)L + \log(N + M + J + D + 2))$$

and bit-operation bound $O(A_{pq} S_{pq}^3)$. This bound includes sorting and comparison. For a fixed root promise, at most $N + 1$ incoming-price labels are reached, independently of the number of local pieces. The behavioral minimization and price-order contiguity conclusions of Theorem EC.4.1 also remain valid.

Proof. On the interior of local piece j , the response is $(r_{vij} - a_{vi}\eta)/q_{vij}$. At a join t , the action equals t for every price between $g'_{vi,+}(t)/a_{vi}$ and $g'_{vi,-}(t)/a_{vi}$. These intervals have the correct orientation by concavity. The two domain endpoints give constant tails. Thus a coordinate with m pieces contributes at most two endpoint events and two events per internal join, or $2m$ in total. At a differentiable join the two events coincide. Strict concavity gives uniqueness, including at a nondifferentiable join; the local response is continuous and nonincreasing.

The suffix-path bijection in Theorem EC.2.1 does not depend on the reward being a single quadratic. Removing the current cap separates coordinates and successors, giving (EC.16). The cap-reflection inequality in that theorem uses only optimality of the pre-cap continuation and linearity of its payment; it therefore applies without change. A weighted sum creates no knot outside the union of its inputs. A cap creates at most one new crossing. Induction over public vertices yields the global event budget. Merging weighted coefficient changes gives (EC.17). Fixed coordinates contribute constants, not knots.

On an open interval between events, local payment sources now have the forms $a_{vi}t_{vi,j}$, $a_{vi}r_{vij}/q_{vij}$, and $-a_{vi}^2/q_{vij}$, together with cap constants. Expanding response coefficients along acyclic paths produces

exactly the same path-product denominator structure as in Theorem EC.2.4. The product of primitive numerators and denominators has logarithm $O(L)$; raising it to $N + O(1)$ bounds the denominator of every path term. Summed absolute path weights times the local sources are bounded by a polynomial in $N + D$ times $2^{O(L)}$. Reduced partial sums therefore have height S_{pq} . Each crossing is a ratio of a constant number of bounded-height coefficients. Exact gcd reduction and cross-product comparison give the stated conservative Turing bound. These conceptual common denominators are not materialized by the implementation.

Only cap barriers change the carried price. Local marginal joins determine an action at that price but do not create a new writable price. Hence the running price is still the maximum of the root price and visited barriers. Along each suffix, every action coordinate is nonincreasing in the initial price; the endpoint-equality argument for contiguous behavioral classes remains applicable. Strict concavity preserves uniqueness and the exact-optimum randomized lower bound. $\square$

The extension includes genuine marginal discontinuities, not only a finer partition of an unchanged quadratic. It does not include affine pieces with zero curvature, coupled promises, or switching costs. Those changes can require different uniqueness or state arguments. The results above extend the exact theorem's domain without using those more general models to support an unproved closure claim.

EC.3.2 A certificate that does not differentiate at a kink

Define the bounded local conjugate

$$k_v(s) = \sum_i \max_{x \in [l_{vi}, h_{vi}]} \{g_{vi}(x) - a_{vi}sx\}.$$

At each reached state (v, η) , let s be its reflected price, $\chi = s - \eta$, and $w_{v,\eta}$ its discounted occupation weight. Then

$$J(b_0) \leq \eta_0 b_0 + \sum_{(v,\eta)} w_{v,\eta} \{\chi B_v + k_v(s)\}. \quad (\text{EC.18})$$

The compiled policy attains equality. Indeed, each local action attains its bounded conjugate, $\chi(B_v - Y_v) = 0$, and successor-price terms telescope under the occupation-flow equations. This is the same dual argument as Section EC.2.4, expressed without a derivative at a join or an artificial choice between adjacent quadratic formulas. All terms are rational at a rational compiled query. Aggregate certificate numerators may be larger than individual response coefficients; the coefficient-height bound is not an assertion that every printed certificate total has that same height.

EC.3.3 Explicit tables versus a shared arithmetic circuit

The lower bound in Theorem EC.2.2 counts pieces in separately materialized vertex-response tables. It is not an information-theoretic lower bound for every exact representation of the family. That distinction leads to a complementary implementation result rather than a weaker size theorem.

Proposition EC.3.2 (Exact shared-circuit storage–query tradeoff). *After compilation, retain the local response curves, graph, cap barriers, and sorted global event set, but discard the per-vertex payment-response*

tables. The resulting shared circuit represents every R_v exactly using $O(J + D + N + M)$ rational scalars. For a common price, all vertex payments can be evaluated in

$$T_{\text{eval}} = O\left(N + M + D + \sum_{vi} \log(m_{vi} + 1)\right)$$

arithmetic operations and comparisons, with the logarithmic sum over nonfixed coordinates. A new feasible root promise can be inverted using $O(T_{\text{eval}} \log(Q + 1))$ such operations. The result is a post-compilation storage bound; it does not assert that the preprocessing peak space is linear. A separately materialized class-transition table remains charged as in Theorem EC.4.1.

Proof. Store one addition-and-cap node per public vertex and one weighted link per edge. Local piecewise-affine evaluators have total size $O(J + D)$. At a supplied price, evaluate the circuit in reverse topological order using (EC.16); shared successors are evaluated once. This gives the storage and evaluation counts. Binary search the globally sorted event locations using the circuit's root payment. Between consecutive events the root response is affine, so evaluations at the two endpoints recover the exact inverse by linear interpolation in the payment–price plane. Constant intervals and tails return any finite supporting price, including a finite event endpoint. A constant root response needs no search. Cap barriers and the global event list have size $O(Q)$. $\square$

For the chain in Theorem EC.2.2, explicit tables have exactly $n^2 + 2n$ segments, whereas the shared circuit has n graph nodes and $3n$ local response segments. Both statements hold simultaneously. Materialization purchases faster repeated inversion; sharing saves read-only representation space. Neither representation changes the extra writable alphabet of an already compiled exact policy. Thus read-only storage, combined execution states, and additional writable symbols are distinct design quantities even within the same problem instance.

EC.3.4 An explicit bridge to parametric flow

The relation to parametric optimization can be stated constructively. In the occurrence tree of Proposition EC.1.1, introduce one vertex per occurrence and a sink. Send a feasible root supply $b \leq B_0$ through the tree. An arc from an occurrence's parent to that occurrence carries its total descendant allocation and has capacity $w_h B_{v(h)}$. An allocation arc from h to the sink carries z_h , has the transformed local bounds, and has cost

$$-w_h g_{v(h)}(z_h / (w_h a_{v(h)})).$$

Tree arcs have zero cost. Flow conservation makes their flows exactly the descendant sums. For multiple local coordinates, use parallel allocation arcs with their corresponding costs and coefficients. Hence this is a value-preserving convex-cost flow representation, including piecewise-quadratic rewards. Lower bounds can be shifted into node balances.

This reduction makes the comparison with Klimm and Warode (2021) precise: root supply is a primal demand parameter, whereas R_0 uses its dual price parameter; inversion connects the two. Applicability

of a particular implementation must account for capacities, lower flows, and zero-cost structural arcs. The occurrence network can still be exponentially larger than the supplied public graph. The asserted contribution is therefore the primitive event budget and exact representation tradeoff on the quotient, not that piecewise-quadratic flow paths are new. The additional exact active-set reference described below computes complete parametric paths on small unfolded problems. It is not presented as an implementation or timing benchmark of the published parametric-flow algorithm.

EC.4 Minimal Additional Writable Memory and Its Contractual Value

EC.4.1 Observation architecture and the reached transducer

Fix the public graph, all primitives, and one root promise. At a decision, the decoder observes the current public vertex, including date, and a writable symbol. A read-only table may contain primitives, response coefficients, thresholds, the query's root price, and transition/output maps. Between decisions the update may depend on the old symbol and the realized public transition; the next action decoder has no separate access to the past transition. A retained promise, past tier, branch identity, or random seed must be charged as writable information if it carries history not contained in the current public vertex. A continuous remaining promise is therefore not an uncharged alternative register.

This architecture deliberately separates three resources. The *additional writable alphabet* is the changing record carried across recombination. The *combined reached behavior states* are the observable public vertex paired with a behavior class. The *read-only representation* contains the graph, compiled responses, coefficient precision, and transition/output tables. None of these quantities is used as an unqualified notion of memory below. Changing the root promise requires a new inversion and can change the reached state set.

Let $\mathcal{P}_v$ be the incoming prices reachable at v under Theorem EC.2.1. For $\eta \in \mathcal{P}_v$, put $s_v(\eta) = \max\{\eta, \alpha_v\}$ with the missing-barrier convention. The reached system is a deterministic output transducer. Its state set is

$$Q = \{(v, \eta) : v \in \mathcal{V}, \eta \in \mathcal{P}_v\}.$$

At state (v, η) the output is $x_v^0(s_v(\eta))$. If the realized public transition is edge (v, w) , the next transducer state is $(w, s_v(\eta))$. The public edge is thus an exogenous input to the deterministic machine.

For two prices at the same public vertex, define behavioral equivalence backward:

$$\eta \sim_v \eta' \iff \begin{cases} x_v^0(s_v(\eta)) = x_v^0(s_v(\eta')), \\ s_v(\eta) \sim_w s_v(\eta') \text{ for every } (v, w) \in \mathcal{E}. \end{cases} \quad (\text{EC.19})$$

At a terminal vertex only the first condition is needed. This is the usual equality-of-output-and-successor-class recursion used in deterministic machine minimization. We use that classical principle explicitly (Bean et al., 1987; Givan et al., 2003; Peyrière, 2023); the new content lies in the contract-generated transducer and its ordered structure.

Theorem EC.4.1 (Minimal reached transducer and additional writable alphabet). *Let c_v be the number of classes of $\sim_v$ on $\mathcal{P}_v$. Under the preceding observation architecture:*

1. *the reached controller has*

$$C_* = \sum_{v \in \mathcal{V}} c_v \quad (\text{EC.20})$$

reachable public-state/class pairs;

2. *the minimum size of a common additional writable alphabet is*

$$K_* = \max_{v \in \mathcal{V}} c_v, \quad (\text{EC.21})$$

hence the changing record needs $\lceil \log_2 K_ \rceil$ bits;*

3. *every class is contiguous in the ordered list of incoming prices at its public vertex;*
4. *the backward signatures can be constructed on the reached vertex–price graph in $O((N+M)N \log(N+1))$ arithmetic comparisons and elementary index operations.*

The read-only response compiler remains charged separately: it stores $O(N^2 + M)$ rational scalars of coefficient height at most S from Theorem EC.2.4. A direct class-transition table has $\sum_v c_v \deg^+(v)$ entries. The exact-optimum lower bound for K_ also holds for randomized controllers whose retained private randomness is charged to the same writable state and whose continuation feasibility is imposed in conditional expectation at each public history.*

Proof. For a finite deterministic output transducer, two states can share one minimized state exactly when they emit the same current output and every common input leads to equivalent successors. Applied within a fixed observable public vertex, this is precisely (EC.19). Backward partition refinement therefore yields the coarsest behavior-preserving partition of the reached price states. This is the classical minimization step.

Under the declared architecture the public vertex is observed without using the writable register. Thus each pair (v, class) is a distinct reached combined state, giving (EC.20), while class indices may be reused across different public vertices. A common alphabet of size $\max_v c_v$ is sufficient: at vertex v , assign distinct labels only to its c_v classes, decode the common action, and update to the corresponding child label. Conversely, if two inequivalent prices at the same public vertex share one writable symbol, the deterministic decoder and update maps coincide from that point onward, so some positive-probability suffix receives a different action than the unique optimum. Hence at least c_v labels are necessary at each v , proving (EC.21).

Along a fixed suffix path, the effective price is the maximum of the incoming price and a fixed finite set of barriers. Every local action is nonincreasing in that effective price. Thus every coordinate of the complete future plan is nonincreasing in the initial incoming price. If two ordered endpoint prices induce the same plan, every intervening price induces that same plan coordinate by coordinate. Hence each equivalence class is contiguous.

A signature consists of the local action and the tuple of successor-class indices. Processing signatures in reverse topological order over at most $N(N+1)$ reached vertex–price pairs and at most $M(N+1)$

pair transitions gives the stated conservative bound. The read-only counts are direct consequences of Theorems EC.2.1 and EC.2.4.

For a randomized feasible controller, take the mean action at each public history. Exogenous transitions and linear payments and caps make this mean vector feasible under conditional-expectation feasibility. Strict concavity gives

$$\sum_h w_h \mathbb{E}_{g_{v(h)}}(X_h) \leq \sum_h w_h g_{v(h)}(\mathbb{E}X_h) \leq J(b_0),$$

with equality only if each action is almost surely constant at the unique optimal value. Distinct required continuation plans must therefore have disjoint supports of retained symbols; a shared symbol with positive probability would impose the same future behavior on both histories. Additional retained randomness capable of distinguishing them is itself writable information under the stipulated accounting. At least c_v symbols are necessary. $\square$

The randomization convention has a concrete contractual interpretation. Conditional-expectation feasibility applies when participation is evaluated before an internal controller coin is realized and the private coin is not a contractually observable service state. If the institution instead requires every private randomization realization to satisfy every continuation cap pathwise, the admissible controller class is smaller. The deterministic optimum and the lower bound above remain valid; no smaller exact alphabet can be created by imposing the stronger feasibility requirement.

Corollary EC.4.2 (Linear writable-memory order is tight). *For every $k \geq 1$, the renewal family in Theorem 4.1 has $N = k + 2$ public vertices and satisfies $K_* = k = N - 2$. Hence the $O(N)$ upper bound on the additional writable alphabet is worst-case tight.*

Proof. The k intermediate branches recombine into one terminal public vertex and require k distinct optimal terminal actions. They are therefore k distinct behavioral classes at that terminal vertex. The root and terminal add two public vertices. $\square$

The theorem minimizes exact execution memory for this fixed optimum under a declared observation interface. It does not minimize the response compiler's storage or characterize the best approximate controller for every alphabet size. Those are separate design questions. The distinction is now explicit in K_* , C_* , and the read-only response size.

EC.5 Mean-Preserving Radial Spreads of Outside Options

Fix positive probabilities, an interior mean $\bar{b}$, and strictly increasing numbers d_j with $\sum_j \pi_j d_j = 0$. Let

$$b_j(\varepsilon) = \bar{b} + \varepsilon d_j, \quad 0 < b_j(\varepsilon) < 1. \quad (\text{EC.22})$$

Keep the common service-benefit continuation fixed. The outside options are that continuation minus $b_j(\varepsilon)$. Increasing ε is therefore a mean-preserving radial spread of continuation caps and outside options along the fixed ordered direction d ; physical public transitions are unchanged.

Proposition EC.5.1 (Mean-preserving radial spreads increase the value of contractual memory). *For the quadratic renewal family in (4), with arbitrary positive branch probabilities and every strictly ordered zero-mean direction d , every insufficient deterministic alphabet $m < k$ has $D_m(\varepsilon) > 0$ and $D_m(\varepsilon)$ is strictly increasing in $\varepsilon > 0$ throughout the admissible range. Moreover $D_m(\varepsilon) = O(\varepsilon)$ as $\varepsilon \downarrow 0$. Exact execution needs one symbol at $\varepsilon = 0$ and k symbols at every $\varepsilon > 0$.*

Proof. For a fixed contiguous partition with cell minimum i , the loss on branch h in that cell is

$$\pi_h \varepsilon (d_h - d_i)(2 - \bar{b} - \varepsilon d_i).$$

For $h > i$, its derivative is

$$\pi_h (d_h - d_i)(2 - \bar{b} - 2\varepsilon d_i) > 0.$$

If $d_i \leq 0$ this is immediate. If $d_i > 0$, admissibility gives $\varepsilon d_i < 1 - \bar{b}$, so the last factor exceeds $\bar{b} > 0$. A partition with fewer than k cells has at least one positive term. Every candidate partition cost is therefore strictly increasing and positive. The minimum of finitely many strictly increasing candidate costs is strictly increasing: evaluate at the minimizing partition for the larger parameter. Each cost is $O(\varepsilon)$ near zero. Distinct optimal terminal tiers give the alphabet statement. $\square$

The proposition is intentionally a theorem about every ordered radial spread direction, not a claim about arbitrary convex-order, majorization, or variance comparisons between unrelated cap vectors. It is nevertheless stronger than an equal-weight or one-calibration statement: branch probabilities are arbitrary positive numbers and the direction d is arbitrary subject to strict order and zero weighted mean. The economic distinction remains sharp: the exact alphabet can jump discontinuously at zero spread while the value loss from insufficient memory converges continuously to zero.

EC.6 The Optimized Shared-Table Comparator

A prescribed observation restriction is optimized before its adaptation value is assessed. Let $c(v) \in \{1, \dots, C_{\text{info}}\}$ be its information cell and let u_c be a single table chosen before uncertainty. The restricted value is denoted K^{tab} , keeping the number of cells distinct from the value.

Proposition EC.6.1 (Zero-release graph formulation). *The optimized zero-release comparator is the convex quadratic program*

$$\begin{aligned} K^{\text{tab}} &= \max_{u, y} \sum_v \bar{w}_v g_v(u_{c(v)}) \\ y_v &= a_v u_{c(v)} + \beta \sum_w p_{vw} y_w, \quad y_v \leq B_v, \quad y_0 = b_0, \\ l_v &\leq u_{c(v)} \leq h_v. \end{aligned} \tag{EC.23}$$

Here $\bar{w}_v = \beta^{t(v)} \Pr(v)$ is the graph's forward reach weight. There are $C_{\text{info}} + N$ variables and $O(N + M + C_{\text{info}})$ coefficient occurrences. For a fixed shared table, an incoming absolute switching cost can also be represented

exactly by its root term and one reach-weighted epigraph term per public edge.

Proof. A common table fixes every local action by the public vertex. Backward conditional expectation gives the displayed y_v at every occurrence of v . Thus all history caps are equivalent to the N displayed caps. Conversely, every feasible table and payment vector implements accepted actions at all histories. Summing rewards by public reach weights proves the objective identity. An edge uses two fixed table entries, so its switching cost is also determined without additional history. $\square$

This supporting formulation uses standard convex optimization; it is not the central algorithmic novelty. In particular, a public table is not the full-information policy class. The difference $J(b_0) - K^{\text{tab}}$ compares two optimized protocols on identical primitives.

At a fixed table and release $\delta \geq 0$, permit $|x_h - u_{c(v(h))}| \leq \delta \omega_{c(v(h))}$, where $\omega_c \geq 0$. Set

$$l_v^{\text{eff}} = \max(l_v, u_{c(v)} - \delta \omega_{c(v)}), \quad h_v^{\text{eff}} = \min(h_v, u_{c(v)} + \delta \omega_{c(v)}).$$

Empty intervals or an infeasible root promise are rejected. Otherwise the finite-price compiler solves this fixed query exactly.

Proposition EC.6.2 (Supporting cuts including coincident bounds). *At every feasible fixed query (u, δ) , physical and corridor multipliers can be split so that nonnegative corridor multipliers $\rho_{v,\eta}^-, \rho_{v,\eta}^+$ give*

$$G_c = \sum_{(v,\eta):c(v)=c} w_{v,\eta} (\rho_{v,\eta}^+ - \rho_{v,\eta}^-), \quad H_\delta = \sum_{(v,\eta)} w_{v,\eta} \omega_{c(v)} (\rho_{v,\eta}^+ + \rho_{v,\eta}^-). \quad (\text{EC.24})$$

For any other feasible design $(\tilde{u}, \tilde{\delta})$,

$$V(\tilde{u}, \tilde{\delta}) \leq V(u, \delta) + G^\top (\tilde{u} - u) + H_\delta (\tilde{\delta} - \delta). \quad (\text{EC.25})$$

This statement includes physical/corridor ties and zero-width intersections.

Proof. At a reached pair let $d = r_v - q_v x - a_v s$. Choose effective lower and upper multipliers $\ell = \max(-d, 0)$, $u^+ = \max(d, 0)$. A positive lower multiplier occurs only when $x = l_v^{\text{eff}}$; a positive upper multiplier only when $x = h_v^{\text{eff}}$. Among the original lower inequalities, at least one attains l_v^{eff} . Assign nonnegative portions summing to ℓ only to the attaining physical and corridor inequalities. Do the analogous split for u^+ . At a tie any such split works. At zero width both lower and upper normal directions are available, and the sign choice still satisfies stationarity; no division by the width is used.

All assigned original multipliers satisfy complementarity because each receiving bound is active. They therefore form full fixed-query Lagrange multipliers with the root price and reflected cap multipliers. Keep them fixed while changing only the corridor right-hand sides. A corridor upper bound contributes $\rho^+(u_c + \delta \omega_c)$ and a lower bound contributes $-\rho^-(u_c - \delta \omega_c)$ to the dual objective. Their change gives (EC.24). Weak duality and exact equality at the anchor prove (EC.25).

The certificate's old price labels merely aggregate the original history multipliers. Those multipliers can be assigned to the same public histories under any candidate policy, because the public process is exogenous. Thus the upper bound applies to every new feasible policy, not just policies sharing the old state labels. $\square$

The oracle supplies a supergradient of the fixed-table value in the design parameters. A continuous Benders-style outer master can use these cuts and feasible-query lower bounds, but no finite exact cut count or global positive-release polynomial algorithm follows. The exact graph-sized zero-release formulation, exact fixed-query compiler, and absence of a finite-cut guarantee for the outer design are distinct results.

EC.7 The Laminar Polymatroid Connection

This section supplies the feasible-set link to Federgruen and Groenevelt (1986); Groenevelt (1991); it is not offered as a new polymatroid theorem. In the variables of Proposition EC.1.1, set $\ell_h = w_h a_{v(h)} l_{v(h)}$, $u_h = z_h - \ell_h$, and $c_h = w_h a_{v(h)} (h_{v(h)} - l_{v(h)})$. The residual subtree capacities are $C_h = w_h B_{v(h)} - \sum_{j \succeq h} \ell_j \geq 0$, and the residual total is $b' = b_0 - \sum_h \ell_h \geq 0$. Write $\mathcal{T}_h$ for the descendant set at h . For $A \subseteq \mathcal{T}_h$, define the rank recursively by

$$f_h(A) = \min \left\{ C_h, c_h \mathbf{1}_{\{h \in A\}} + \sum_{j \text{ child of } h} f_j(A \cap \mathcal{T}_j) \right\}. \quad (\text{EC.26})$$

The root function f_0 is normalized, nondecreasing, and submodular. Indeed, the expression inside the minimum is a sum of a nonnegative modular function and submodular functions on disjoint ground sets; truncation at a nonnegative constant preserves nondecreasing submodularity. The latter fact follows directly from diminishing marginal increments: the increment of a truncated function is its original nonnegative increment capped by the remaining distance to the truncation level, both of which cannot increase when the conditioning set grows.

The residual packing region is exactly $P(f_0) = \{u \geq 0 : \sum_{h \in A} u_h \leq f_0(A) \text{ for every } A\}$. To verify the rank, maximize allocation to a subset A on a subtree. The independent child capacities contribute their ranks and the current coordinate contributes $c_h \mathbf{1}_{\{h \in A\}}$; the current cap truncates their sum. Any value between zero and that sum can be obtained by scaling a feasible nonnegative allocation, which proves (EC.26). Conversely, the subset inequalities for singleton coordinates and for descendant sets imply the original box and subtree rows, since their ranks do not exceed the respective capacities. This proves equality of the regions.

Feasibility gives $b' \leq f_0(\mathcal{T})$. Define $f^{b'}(A) = \min\{b', f_0(A)\}$. The exact-payment feasible set is the base

$$\left\{ u \in P(f^{b'}) : \sum_h u_h = f^{b'}(\mathcal{T}) = b' \right\}.$$

Truncation matters when the promised total is below the maximum feasible allocation. The objective after translation remains separable strictly concave. Hence both the laminar and the polymatroid connections are exact, including nonzero local lower bounds and a root equality. Our input-compression and execution-memory claims concern this repeated-subtree subclass, not separable allocation over all polymatroids.

EC.8 Coefficient-Event Compilation

For each response, store ordered rational knots $k_1 < \dots < k_s$ and one affine coefficient pair (m_i, c_i) on each interval, including the two tails. Equal adjacent pairs are removed. Evaluation uses binary search

and the convention that a knot belongs to its right interval; continuity makes the value independent of that convention.

At vertex v , initialize the left-tail coefficient with $a_v h_v + \beta \sum_w p_{vw} U_w$. The local clipped payment changes slope and intercept at

$$(r_v - q_v h_v)/a_v, \quad (r_v - q_v l_v)/a_v.$$

For each successor knot, add the weighted difference between its new and old coefficient pair to the event at that knot. Combining equal-key events before sorting ensures that a repeated downstream breakpoint is stored once at the current vertex. Starting from the left-tail pair, sweep events to recover the pre-cap lines. This operation merges coefficients; it never recovers a slope from two evaluations at newly generated cap crossings.

If the cap is redundant, store the pre-cap curve. Otherwise find the first crossing of B_v by scanning the nonconstant affine pieces. Replace the entire prefix by the constant B_v , retain the original crossing line to its right, and discard any newly equal neighboring pairs. The crossing may coincide with an existing knot or lie at the left endpoint of a constant equality plateau. Both cases use the same continuity convention.

The input checker requires rational integers or strings, positive payment and quadratic coefficients, a valid physical interval, positive outgoing probabilities summing to one, topological vertex order, and reachability from the root. Duplicate edges must first be aggregated. It rejects an empty feasible continuation interval or an out-of-range root promise rather than returning an approximate feasible answer. Floating-point primitives are rejected by the exact compiler because their intended rational interpretation would otherwise be ambiguous.

EC.8.1 A direct plateau argument

Let x_1, x_2 maximize $G(x) - \eta_1 Y(x)$ and $G(x) - \eta_2 Y(x)$ over the same compact convex pre-cap set, with $\eta_1 < \eta_2$. Optimality gives

$$G(x_1) - G(x_2) \geq \eta_1 (Y(x_1) - Y(x_2)), \quad G(x_1) - G(x_2) \leq \eta_2 (Y(x_1) - Y(x_2)).$$

If the two payments coincide, these inequalities imply equal rewards. Both actions then maximize both objectives. Strict concavity forces $x_1 = x_2$. A payment plateau therefore changes neither the optimal local tier nor any future tier, although the price supporting that plan may vary. This proves policy uniqueness at a cap-equality plateau without assuming uniqueness of a Lagrange multiplier.

At a response's upper tail, the implementation returns one finite price strictly below its first knot. At the lower tail, it returns an endpoint of the last decreasing segment; a completely constant response uses price zero. None of these choices changes the optimum. If the root itself is capped exactly at its promised payment, its cap multiplier and equality price may be redistributed, but their contribution to the objective remains the same because the two right-hand sides agree.

EC.8.2 Explicit pseudocode

The following procedure is equivalent to the executable coefficient-event implementation.

Process vertices in reverse topological order. At each vertex, validate the feasible lower payment. Create the two local clipping events when the interval is nondegenerate. Add discounted probability-weighted successor coefficient changes into a common event dictionary. Sort its keys and sweep to obtain the pre-cap curve. When necessary, invert that curve at the current cap and replace its left prefix by a constant. Store the remaining lines and current barrier. After the root query arrives, invert the root curve at the exact promised payment. Propagate the maximum of the incoming price and current barrier, aggregate identical reached vertex–price pairs, and record actions and multipliers. Finally minimize the reached execution graph by backward behavioral signatures.

The procedure constructs all graph response curves, but it constructs execution states only for the supplied query. A different query may change both the reached pairs and the minimized machine. The table-storage bound and writable-alphabet bound describe different data structures.

EC.9 Bit Bounds for Execution and Certificates

The main bit theorem deliberately concerns response preprocessing and one inversion. The next observation explains why exact execution remains polynomial while a printed total value can be much longer than an individual response coefficient.

Proposition EC.9.1 (A conservative execution-height bound). *Let S be the bit-height bound in Theorem EC.2.4, with the root query included in the input length. For a fixed query, all rational numbers needed by a compact execution certificate, including its aggregate reward and dual value, have bit height $O(NS + NL + \log(N + 1))$ under a common-denominator implementation. Exact forward/backward execution and auditing thus admit polynomial bit complexity on at most $N(N + 1)$ reached pairs and $M(N + 1)$ transitions.*

Proof. The discounted weight of a reached pair is a sum of probabilities of public paths that produce its price label. Each path contains at most $N - 1$ edges. The same product-of-input-denominators argument as in Theorem EC.2.4 bounds its rational denominator by D_0^{N+4} and its numerator by a comparable height, since its weight is at most one. This representation does not multiply the denominators of individual histories.

There are at most $N + 1$ distinct numerical prices. A local action is either a physical endpoint or $(r_v - a_v s)/q_v$ for one such price s . A local reward has degree at most two in that price, with rational primitive coefficients. A common denominator for all weighted reward contributions divides a fixed power of D_0^{N+4} times the product of the squared denominators of the distinct prices. Its logarithm is $O(NL + NS)$. The same denominator argument applies to payments, cap-multiplier contributions, bound multipliers, and the squared local dual terms; a difference of two prices introduces denominators already in this product. The sum contains only polynomially many compact-pair terms, and their magnitudes add at most polynomial factors and the bounded-height price magnitudes. This gives the stated height. Forward and backward dynamic programming on the reached graph then uses polynomially many operations on polynomial-height integers. $\square$

This is a conservative implementation bound, not an assertion that the experimental total rewards attain it. In the implementation, reward and payment contributions are first grouped by effective price before their final sum. This reduces repeated greatest-common-divisor work across unrelated price denominators without changing the exact result. The independent audit reconstructs conditional payments directly from the action/transition records and checks local normal-cone conditions; it does not trust the compiler’s claimed response identity.

For trusted internally generated experiments, the worker explicitly lifts Python’s default decimal integer-string length guard. This affects serialization only, not mathematical arithmetic or input validity. The general compiler does not silently reinterpret floating-point inputs or accept arbitrary external executable data. The worker’s total reward can have many more decimal digits than a stored slope, precisely the distinction addressed by the proposition.

EC.10 Behavioral Quotients and Information Accounting

A backward signature at a reached pair consists of its rational action and the ordered tuple of successor-vertex and successor-class indices. Two signatures are equal precisely when the corresponding complete suffix plans are equal. Because the graph is acyclic, successor classes are already known when a vertex is processed. The stored class index may be reused at other public vertices. The update map has the public transition as input and writes the successor’s local class before the next action is decoded.

This timing convention is important. A past branch cannot be made available to the next action decoder as an additional uncharged register merely by calling it the last observed edge. Similarly, a remaining promise, past service tier, or retained seed would enlarge the observation model if made separately available. The minimal-alphabet theorem fixes these interfaces. It compares controllers with the same read-only program and current public observation, not architectures with different free history channels.

EC.10.1 A strict reduction beyond counting numerical prices

Take a root fixed at zero, two equally likely intermediate vertices, and a common terminal vertex fixed at tier $1/4$. Intermediate rewards are $2x - x^2/2$, their caps are $1/2$ and 1 , payments and discounting are one, and the root promise is $3/4$. Both branch caps bind. The intermediate actions are $1/4$ and $3/4$, so two different prices can reach the terminal vertex. Yet its terminal action is fixed and has no successors. Both prices therefore belong to the same terminal behavioral class, and the entire optimal protocol can use one writable symbol. The exact test records two raw states at that vertex but a minimum alphabet of one. Counting numerical prices alone would overstate necessary memory.

EC.10.2 Randomization and restricted-memory values

The Jensen argument in Theorem EC.4.1 concerns a randomized controller attaining the *full* optimum. It forces every action at every positive-probability public history to equal the unique optimum almost surely. It does not prove that randomization leaves all lower-alphabet optima unchanged. The frontier in Theorem 4.1

is consequently stated for deterministic encodings. These two results have different conclusions and should not be merged into an unsupported randomized-frontier claim.

EC.11 Heterogeneous Cell Costs and Dispersion

For quadratic terminal reward $f(c) = rc - qc^2/2$ and intermediate cost $h_h(y) = \gamma_h y^2/2$, the cell cost at minimum b_i expands as

$$C(i, j) = \sum_{h=i}^j \pi_h \{rb_h - qb_h^2/2 - rb_i + qb_i^2/2 + \gamma_h b_h^2/2 - \gamma_h b_h b_i + \gamma_h b_i^2/2\}.$$

Prefix sums of π_h , $\pi_h b_h$, $\pi_h b_h^2$, $\pi_h \gamma_h$, $\pi_h \gamma_h b_h$, and $\pi_h \gamma_h b_h^2$ therefore evaluate a cell in constant arithmetic time. With general convex cost functions, the segmentation recurrence still applies after cell costs are computed; the quadratic prefix-sum preprocessing bound is not assigned to an arbitrary function oracle.

The finite enumeration test does not assume contiguous cells. It enumerates all set partitions for branch counts two through seven, computes each cell at its smallest cap, and compares the best value at every cell count with the ordered dynamic program. Two cost families are used: heterogeneous quadratic costs with quadratic terminal reward, and heterogeneous cubic costs with a strictly increasing piecewise-concave terminal reward. Probabilities are proportional to branch indices rather than equal. In total, 2,308 arbitrary partitions are checked exactly.

For the dispersion test, probabilities are again unequal, the mean cap is $1/2$, and centered branch indices define d_j . Four rational positive dispersion levels are tested for each branch count. The resulting 84 strict loss comparisons complement, but do not establish, the analytic monotonicity proof. The limiting alphabet jump at zero dispersion follows from the optimal terminal actions, not from a numerical precision threshold.

EC.12 Complete Endpoint-Normal Splitting

The effective interval is the intersection of four half-spaces on one coordinate: physical lower and upper bounds and corridor lower and upper bounds. At a lower endpoint, its outward lower normal is generated by any original lower inequality attaining the maximum lower bound; at an upper endpoint, the upper normal is generated by an inequality attaining the minimum upper bound. At a singleton, both normal directions are present, even if supplied by different original inequalities. Thus their nonnegative sum is the entire required interval normal cone.

For example, let the physical upper bound be $1/2$ and let the corridor upper bound also equal $1/2$. A positive effective upper multiplier u^+ can be split as θu^+ to the corridor and $(1 - \theta)u^+$ to the physical bound for any $0 \leq \theta \leq 1$. Both original constraints are active, so stationarity and complementarity persist. The resulting supporting cut need not be unique. At zero corridor width, both corridor endpoints coincide; the residual's sign determines a valid orientation without dividing by that width. A physical upper bound

coinciding with a corridor lower bound similarly creates a singleton, and the corresponding two opposite normals span every residual sign.

The boundary suite tests physical-first, corridor-first, and equal splitting. It includes ordinary coupled-graph queries, upper and lower physical ties, fixed physical tiers, zero corridor widths, and intersections reduced to one point by different physical and corridor bounds. For each feasible anchor, nearby rational table/width queries are solved and the proposed upper support is evaluated exactly. Infeasible test queries are rejected rather than counted as supporting-cut successes. The recorded 150 anchors and 1,185 inequalities are over feasible queries only.

EC.13 Read-Only, Combined-State, and Writable Accounting

The residual contribution can be audited without assigning novelty to the classical reductions on which it builds. Table H.3 separates the inherited principle from the model-specific statement and records whether the relevant size bound is matched by a worst-case construction.

The table is deliberately asymmetric. Generic parametricity, normalized memoization, deterministic machine minimization, and finite policy graphs are predecessor principles. The paper’s theorem burden is instead on the compact-input quotient, sharp response size, exact binary implementation, and the contractual interpretation of the minimized reached machine.

Proposition EC.13.1 (Three-layer representation accounting). *Under the assumptions of Sections EC.1–EC.4, the exact representation separates into the following resources.*

1. *Compilation. The complete read-only response has $O(N)$ distinct global knots, $O(N^2 + M)$ stored rational scalars, and polynomial coefficient height and preprocessing bit complexity. The knot and scalar-storage orders are worst-case tight when $M = O(N)$.*
2. *Fixed-query execution. A root promise reaches at most $N + 1$ raw incoming-price labels. After classical behavioral minimization, the combined reached state count is $C_* = \sum_v c_v$, while the minimum additional writable alphabet under the declared public-observation interface is $K_* = \max_v c_v = O(N)$. The $O(N)$ writable-alphabet order is worst-case tight.*
3. *Query reuse. Once the complete response is compiled, a new feasible root promise is inverted with $O(\log(N + 1))$ exact comparisons plus a constant number of rational arithmetic operations; constructing and auditing the reached execution machine for that query remains separately charged.*

These quantities are not interchangeable: K_ is not total controller-state complexity, C_* is not read-only compiler storage, and the root-inversion bound does not include policy emission or certification.*

Proof. Part 1 combines Theorems EC.2.1, EC.2.2, and EC.2.4. Part 2 combines the fixed-query label bound in Theorem EC.2.1, Theorem EC.4.1, and Corollary EC.4.2. Part 3 is Corollary EC.2.5. The final distinctions follow from the observation architecture in Section EC.4. $\square$

This accounting also fixes the interpretation of the service result. A public vertex is an observed operating condition; the writable symbol records only history that is not already present in that public condition. If a different implementation makes additional inherited variables freely observable, the behavioral quotient must be recomputed under that interface. Conversely, charging the public state itself to the controller enlarges total state storage but does not alter the conditional lower bound on the extra writable alphabet established here.

EC.14 Global Codebook Optimization: Implementation and Verification

This section documents Theorems 6.2 and 6.4. The complete structural and algorithmic proofs appear in the article; no omitted numerical experiment is being used to supply a proof step.

EC.14.1 Rational implementation concordance

This subsection documents the retained quadratic enumeration engine. The current linear engine uses the same rational primitives but reconstructs only the winning codebook once per budget.

The implementation accepts integers, rational strings, or exact fractions. It rejects binary floating-point inputs, unordered or repeated caps, nonpositive probabilities, probabilities not summing to one, negative intermediate curvature, and terminal parameters outside $r \geq q > 0$. Sorted distinct caps are part of the interface; sorting unorganized observations would first require sorting their associated probabilities and costs together. An arbitrary-order interface would add $O(k \log k)$ rational comparisons.

The six prefix moment arrays evaluate (9) and (11) without summing over an interval at each query. The routine `tail_polynomial` returns the exact coefficients A, B, C of $Az^2 + Bz + C$; `tail` compares all clamped cell minima and saves the attaining final level. The routine `randomized_frontiers` implements (10) using two cost rows and retained predecessor indices, then combines them with terminal costs. A strict improvement stores a reconstructed codebook; ties can keep any incumbent. Duplicate terminal levels are deleted. Budgets greater than the number of branches repeat the zero-loss solution.

The bit bound does not require that the implementation form a common denominator. Every addition, multiplication, and division returns a reduced fraction. Comparisons are exact cross products. Moment arrays contain sums of fixed-degree input products. Edges use products of moment values and cap endpoints. Tail minimizers divide two such coefficients, with a strictly positive divisor on nonterminal cells. Prefix paths add at most k edge costs, and a complete candidate adds one optimized tail. This dependency pattern rules out uncontrolled nested algebraic extensions: all candidates and the reconstructed lottery probabilities are rational. Evaluating final probabilities adds differences and a division of rational codewords and has the same polynomial height bound. The stated conservative S covers both objective values and reconstructed controls, rather than only response coefficients.

For convex nondecreasing intermediate costs that are not quadratic, Lemma 6.1 and the prefix recurrence still apply when terminal reward is quadratic. Each terminal interval instead requires minimizing a strictly convex one-dimensional function. Thus the finite reduction persists with a one-dimensional convex

minimization oracle. Exact rational attainment and the particular Turing bound are not transferred to a cost family without an exact representation and such an oracle.

EC.15 The Retained Divide-and-Conquer Alternative

The routine `monotone_minima` processes a middle row, scans only the intersection of its feasible columns with inherited lower and upper bounds, and selects the first attaining column. It recurses to the left with that column as upper bound, and to the right with it as lower bound. Induction on the recursion tree shows that the true leftmost row minimizer remains in every search interval. At one depth the intervals have disjoint interiors, with at most one duplicated boundary per sibling pair. The sum of their lengths is $O(k + 2^d)$ at depth d . Summing over $O(\log(k + 1))$ depths gives $O(k \log(k + 1))$ calls, with $O(\log(k + 1))$ recursion stack.

The randomized routine first obtains all terminal minima. It then builds the anchored prefix layers, using the single reachable endpoint at the first layer rather than querying infeasible entries. It retains two cost rows and one predecessor for each reachable layer/endpoint pair. Combination with the terminal costs is a linear scan per budget. Only the winning candidate is backtracked, once per budget. This detail prevents an uncharged reconstruction loop from being hidden inside the improved search bound. Codebooks share no uncharged history register: the reconstructed symbols and their probabilities use the same decoder interface as the unaccelerated algorithm.

The deterministic routine uses the analogous prefix search with the quadratic cell cost evaluated by the same six prefix moments. The structural cell Monge inequality is valid more generally, but a constant-work cell oracle is part of the stated complexity interface. Sorting unordered inputs adds $O(k \log k)$ exact comparisons and must carry probabilities and costs with each cap. The implemented interface instead rejects repeated or unsorted caps. It also rejects floating inputs, invalid probability sums, negative curvatures, and noninteger or nonpositive alphabet budgets.

All economic calculations use reduced fractions. Neither search introduces an additional optimization parameter or nested algebraic extension. Terminal minimizers still require only the quadratic coefficient ratio and exact clamping; prefix costs still add at most k edges. Thus the height bound in Theorem 6.2 remains a valid conservative bound for values, controls, comparisons, and reconstructed lottery probabilities. Oracle-call counts in the computational table are not represented as bit-operation counts.

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